Analysis and Design of RLC Circuits as Resonators and Filters using Discrete-Time State-Space Models

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***Abstract*—This report details the simulation and analysis of discrete-time models for RC, RL, and RLC circuits. The primary objective is to derive and implement a state-space model for a series RLC circuit to analyze its response to step and sinusoidal inputs, and to design audio filters for a final competition. The system is modeled using the Forward Euler approximation with a fixed time step of $h = 1/(192\,\mathrm{kHz})$ [1]. The state-space matrices (A, B) for the RLC circuit are derived from first principles and provided equations. Analysis of the step response revealed the critical nature of numerical stability, where the simulation is only stable for a resistance $R > h/C$; this was demonstrated by comparing an overdamped ($1\,\mathrm{k}\Omega$), a stable underdamped ($55\,\Omega$), and a numerically completely unstable ($0.5\,\Omega$) case. Sinusoidal analysis confirmed that taking the output voltage across the resistor (V_R) configures the circuit as a bandpass filter. This model was then used to design three competition-grade filters: 1) a $440\,\mathrm{Hz}$ "tuning fork" resonator, 2) an $84\,\mathrm{Hz}$ sensor to isolate a helicopter signal from noise, and 3) a 200-$2000\,\mathrm{Hz}$ music filter that successfully removed $60\,\mathrm{Hz}$ hum and high-frequency hiss from an audio file.**

I. Introduction

Resistor-Inductor-Capacitor (RLC) circuits are fundamental components in electrical engineering, forming the basis for oscillators, tuning circuits, and filters. Understanding their multifaceted behavior is essential. While their continuous-time behavior is well-described by differential equations, modern simulation relies on discrete-time models. This case study explores the implementation of such models in MATLAB.

The challenge lies in accurately representing the continuous-time physics of capacitors and inductors using discrete-time difference equations. We employ the Forward Euler method, a first-order numerical approximation, to model the system's evolution over small time steps h [1]. This approach transforms the circuit's differential equations into a linear dynamical system, which can be expressed in the form: $\mathbf{x}_{k+1} = A\mathbf{x}_k + B\mathbf{u}_k$.

The goals of this project are multifaceted and fourfold: 1) to implement and analyze discrete-time models for basic RC and RL circuits; 2) to derive the state-space matrices A and B for a series RLC circuit from its governing physical equations; 3) to simulate and analyze the RLC circuit's step and sinusoidal responses, investigating stability; and 4) to design and implement three tuned RLC circuits for a competition [1]. This report details the derivation of the model, analyzes the simulation results, and provides a detailed justification for the design choices made in the competition.

II. Simulation Methodology

All simulations were conducted in MATLAB. For Parts 1 and 2, the sampling interval h was chosen relative to the circuit's time constant τ to demonstrate numerical accuracy. For Part 3 and the Part 4 competition, a universal high-resolution sampling interval of $h = 1/(192\,\mathrm{kHz}) \approx 5.2\,\mu\mathrm{s}$ was used, as required by the case study instructions [1]. This small step size is crucial for accurately modeling high-frequency audio signals and for maintaining numerical stability in the RLC simulations.

III. Part 1: RC Circuit Analysis

A. Task 1.1: RC Step Response

A discrete-time model for the series RC circuit (Fig. 1 in [1]) was implemented. With $R = 1\,\mathrm{k}\Omega$ and $C = 1\,\mu\mathrm{F}$, the time constant is $\tau = RC = 1\,\mathrm{ms}$. A "good" time step of $h = 0.1\tau = 0.1\,\mathrm{ms}$ was chosen. The circuit's response to a $1\,\mathrm{V}$ step input is shown in Fig. 1, which correctly depicts the capacitor's voltage V_C asymptotically approaching the $1\,\mathrm{V}$ input.

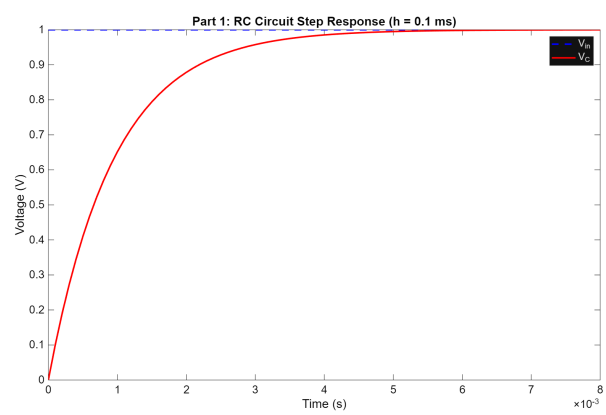


Fig. 1: RC circuit response to a $1\,\mathrm{V}$ step input, simulated with an accurate time step $h = 0.1\,\mathrm{ms}$.

B. Task 1.2: Simulation Accuracy vs. Time Step

To analyze the effect of h on accuracy, the simulation was re-run with an "inaccurate" time step $h_{bad} = 1.5\tau = 1.5\,\text{ms}$. The Forward Euler update for this circuit is stable only for $h < 2\tau$. Although h_{bad} is technically stable, it is important to differentiate stability (the simulation does not blow up) from accuracy (how close the discrete solution is to the continuous-time one). A large h can be stable but, as shown here, highly inaccurate.

Fig. 2 compares the accurate simulation, the inaccurate simulation, and the theoretical curve $V_C(t) = 1 - \exp(-t/\tau)$. The inaccurate simulation clearly overshoots the target voltage and oscillates before converging, a non-physical artifact of the large time step. This confirms that the behavior of the *real* capacitor does not change, but our simulation's ability to model it is critically dependent on h.

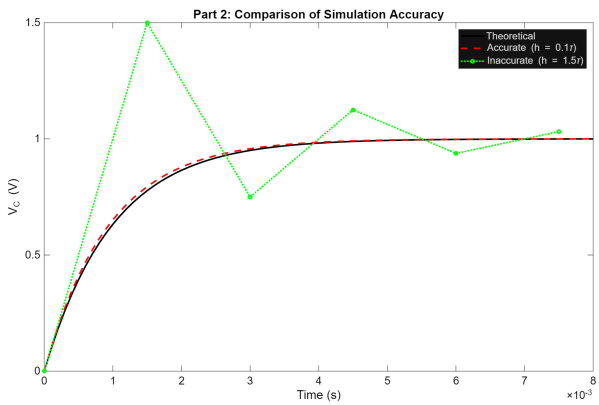


Fig. 2: Comparison of RC circuit simulations. The "inaccurate" simulation ($h = 1.5\tau$) oscillates and overshoots, failing to capture the true physical behavior shown by the theoretical curve.

C. Task 1.3: The RC Time Constant

The time constant $\tau = RC$ represents the time required for the capacitor to charge to $1 - \exp(-1) \approx 63.2\%$ of its final voltage. As seen in the plots, it dictates the "speed" of the circuit's response. After 5 time constants (5τ), the capacitor is charged $1 - \exp(-5) \approx 99.3\%$, which is often considered fully charged.

IV. Part 2: RL Circuit Analysis

A. Task 2.1: RL Step Response

A similar model was implemented for the series RL circuit (Fig. 3 in [1]) with $R = 100\,\Omega$ and $L = 100\,\text{mH}$, giving a time constant $\tau = L/R = 1\,\text{ms}$. Using $h = 0.1\,\text{ms}$, the response to a $1\,\text{V}$ step was simulated. Fig. 3 shows the voltage across the inductor, V_L. At $t = 0$, the inductor opposes the change in current, so all $1\,\text{V}$ is across V_L. As current builds, V_L decays to $0\,\text{V}$.

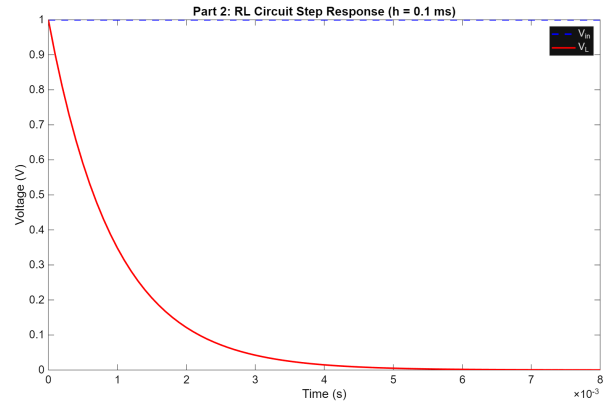


Fig. 3: RL circuit response to a $1\,\text{V}$ step input. The voltage across the inductor, V_L, starts at $1\,\text{V}$ and decays to $0\,\text{V}$ as current reaches a steady state.

B. Task 2.2: Complementary Characteristics

The simulation data from Parts 1 and 2 demonstrate the complementary nature of capacitors and inductors in a DC steady state (SS). The final values of our simulations are summarized in Table I.

TABLE I: Steady-State (SS) Values for RC and RL Circuits

Component	SS Voltage	SS Current
Capacitor (V_C, I_C)	$V_{C,ss} \approx 1.00\,\text{V}$	$I_{C,ss} \approx 0.00\,\text{A}$
Inductor (V_L, I_L)	$V_{L,ss} \approx 0.00\,\text{V}$	$I_{L,ss} = 0.01\,\text{A}$

- **Capacitor:** In SS, V_C matches the $1\,\text{V}$ source. No voltage remains across the resistor ($V_R = V_{in} - V_C = 0$), so the current I_C drops to $0\,\text{A}$. The capacitor acts as an **open circuit**.
- **Inductor:** In SS, V_L decays to $0\,\text{V}$. The full $1\,\text{V}$ source voltage is dropped across the resistor, and the current I_L is limited only by R: $I_L = V_{in}/R = 1\,\text{V}/100\,\Omega = 0.01\,\text{A}$. The inductor acts as a **short circuit** (a wire).

V. Part 3: RLC Circuit Analysis

A. RLC State-Space Model Derivation (Task 3.1)

The core of this project is the model for the series RLC circuit (Figure 5 in [1]). We define the state vector as $\mathbf{x}_k = [v_{C,k}, i_k]^T$. The input is $u_k = v_{in,k}$. The state-update equation $\mathbf{x}_{k+1} = A\mathbf{x}_k + B\mathbf{u}_k$ is derived from two core equations.

1. Capacitor Voltage Update: From $i_k = C\frac{v_{C,k+1} - v_{C,k}}{h}$:

$$v_{C,k+1} = (1)v_{C,k} + (\frac{h}{C})i_k + (0)u_k \tag{1}$$

2. Inductor Current Update: From KVL, $v_{in,k} - v_{L,k} - v_{C,k} - v_{R,k} = 0$. Substituting $v_{L,k} = L\frac{i_{k+1} - i_k}{h}$ and $v_{R,k} = i_k R$ and solving for i_{k+1} yields:

$$i_{k+1} = (-\frac{h}{L})v_{C,k} + (1 - \frac{hR}{L})i_k + (\frac{h}{L})u_k \tag{2}$$

3. Final Matrices: Combining (1) and (2) gives the final state-space model:

$$A = \begin{bmatrix} 1 & h/C \\ -h/L & 1 - hR/L \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ h/L \end{bmatrix} \tag{3}$$

These matrices were implemented in our MATLAB function ‘simRLC‘.

B. Step Response and Stability (Task 3.2)

We simulated the response to a 1 V step input with $L = 10\,\text{mH}$ and $C = 0.1\,\mu\text{F}$. The physical circuit rings if $R < 2\sqrt{L/C} \approx 632.5\,\Omega$ [2]. However, the Forward Euler method has its own stability constraint. With $h = 1/(192\,\text{kHz})$, the stability limit is:

$$R > \frac{h}{C} = \frac{1/192000}{0.1 \times 10^{-6}\,\text{F}} \approx 52.1\,\Omega \tag{4}$$

The discrete-time state matrix's characteristic polynomial is $\lambda^2 - (2 - \frac{hR}{L})\lambda + (1 - \frac{hR}{L} + \frac{h^2}{LC}) = 0$. Applying Schur stability conditions (roots inside the unit circle) reduces to $1 - \det(A) > 0$, which simplifies to $R > h/C$. This is a property of the numerical integrator; the *physical* circuit is stable for all $R \geq 0$.

We tested three cases, as shown in Fig. 4:

- **Quick Decay (Overdamped):** $R = 1000\,\Omega$. This is $> 632.5\,\Omega$ and sounds like a dull "click."
- **Slow Decay (Underdamped):** $R = 55\,\Omega$. Since $52.1\,\Omega < R < 632.5\,\Omega$, this is numerically stable and physically underdamped. It sounds like a high-pitched "ping."
- **Unstable:** $R = 0.5\,\Omega$. Since $R < 52.1\,\Omega$, the simulation is numerically unstable. It sounds like a "pop" as the values blow up to ‘Inf‘.

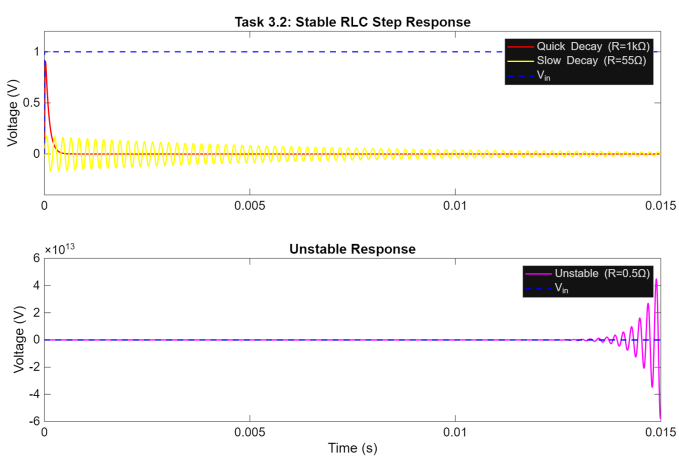


Fig. 4: Step response of the RLC circuit ($V_{out} = V_R$) for three cases. (Top) Stable responses for Overdamped ($R = 1\,\text{k}\Omega$) and Underdamped ($R = 55\,\Omega$) cases. (Bottom) Numerically unstable response ($R = 0.5\,\Omega$).

C. Sinusoidal Response (Task 3.3)

We analyzed the steady-state response using $R = 100\,\Omega$, $L = 100\,\text{mH}$, and $C = 0.1\,\mu\text{F}$. The resonant frequency is $f_0 = 1/(2\pi\sqrt{LC}) \approx 1591.5\,\text{Hz}$. As shown in Fig. 5, the output V_R is maximized at the resonant frequency and attenuated at low (100 Hz) and high (10 kHz) frequencies. This is the definition of a bandpass filter [3]. When listening, the volume corresponds to the amplitude, and the frequencies are audible as low, medium, and high pitches.

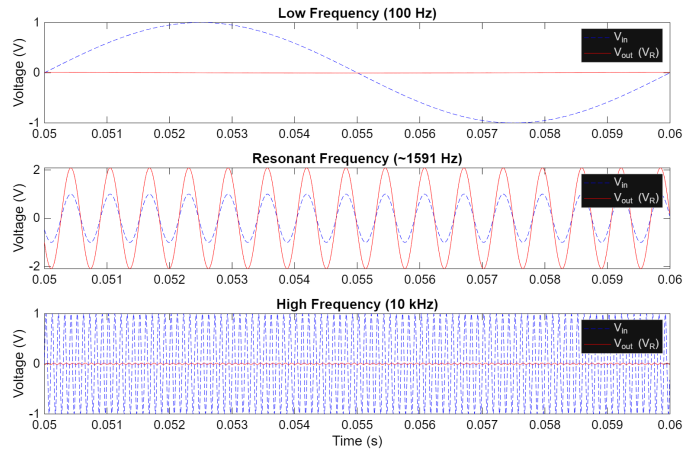


Fig. 5: Sinusoidal response at three frequencies. (Top) At $f = 100\,\text{Hz}$, output is small. (Middle) At resonant $f \approx 1591\,\text{Hz}$, output amplitude equals input. (Bottom) At $f = 10\,\text{kHz}$, output is small. This demonstrates bandpass filter behavior.

VI. Part 4: Competition Filter Design

For a series RLC circuit, the resonant frequency f_0, quality factor Q, and -3dB bandwidth Δf are given by:

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad f_0 = \frac{\omega_0}{2\pi}, \quad Q = \frac{\omega_0 L}{R}, \quad \Delta f = \frac{f_0}{Q} = \frac{R}{2\pi L} \tag{5}$$

These formulas justify the choice of R for controlling either the "ringiness" (high Q, low Δf) or the passband width (low Q, high Δf). The parameters for our competition designs are summarized in Table II.

TABLE II: Competition Circuit Design Parameters

Parameter	Resonator	Sensor	Music Filter
f_0 (target)	440 Hz	84 Hz	≈ 632 Hz
L	≈ 0.13 H	≈ 3.6 H	≈ 0.63 H
C	1.0 µF	1.0 µF	0.1 µF
R	$\approx 5.2 \times 10^{-3}\,\Omega$	$150\,\Omega$	$\approx 7.1\,\text{k}\Omega$
Q (calc.)	$\approx 69,000$	≈ 12.7	≈ 0.35
Δf (calc.)	≈ 0.006 Hz	≈ 6.6 Hz	≈ 1800 Hz

A. Task 4.1: Tuning Fork Resonator

Goal: Create a circuit that rings at $f_0 = 440\,\text{Hz}$ (A_4).

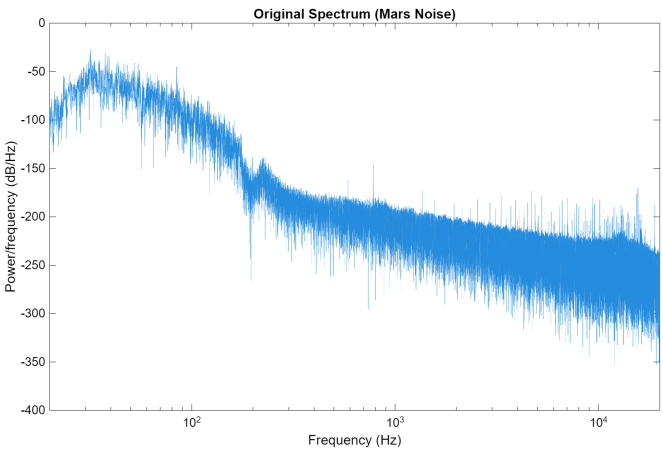
Design Justification: We set $f_0 = 440\,\text{Hz}$ and chose $C = 1.0\,\mu\text{F}$, solving for $L \approx 0.13\,\text{H}$. To maximize ring time

(maximize Q), we must minimize damping, R, by setting it just above the stability limit: $R = 1.001 \times (h/C) \approx 5.2 \times 10^{-3}\,\Omega$. A single $1\,\mathrm{V}$ pulse was sufficient to make it ring audibly.

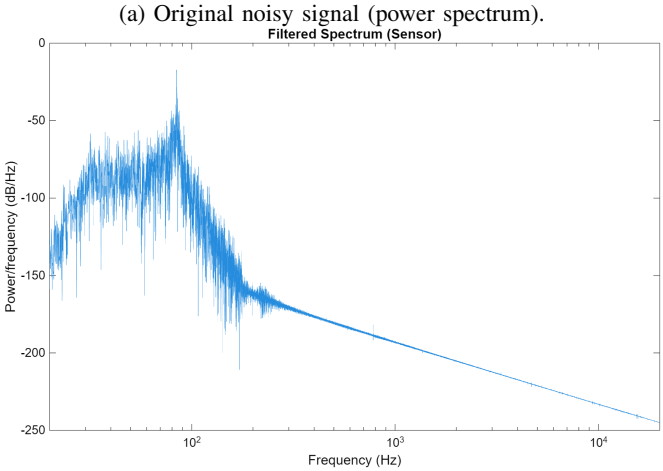
B. Task 4.2: Mars Audio Sensor

Goal: Isolate an $84\,\mathrm{Hz}$ helicopter rotor sound.

Design Justification: This is a bandpass filter centered at $f_0 = 84\,\mathrm{Hz}$. We set $C = 1.0\,\mu\mathrm{F}$ and solve for $L \approx 3.6\,\mathrm{H}$. A medium value of $R = 150\,\Omega$ was chosen to create a narrow passband that rejects wind noise without "ringing" excessively. **Result:** As shown in Fig. 6, the filter successfully eliminated broadband wind noise, isolating the $84\,\mathrm{Hz}$ signal.



(a) Original noisy signal (power spectrum).



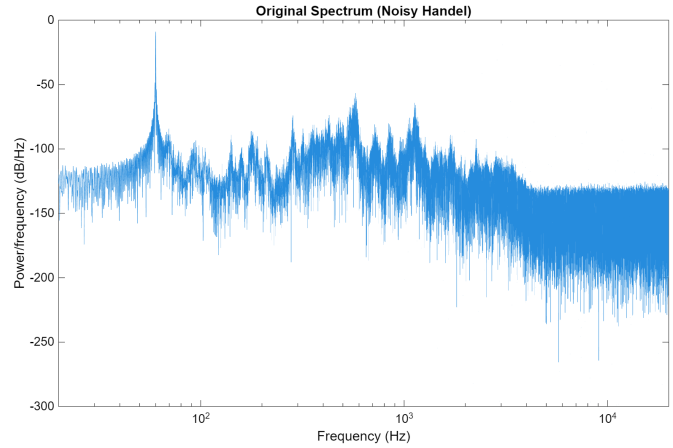
(b) Filtered signal (power spectrum).

Fig. 6: Power spectra for Mars Audio Sensor.

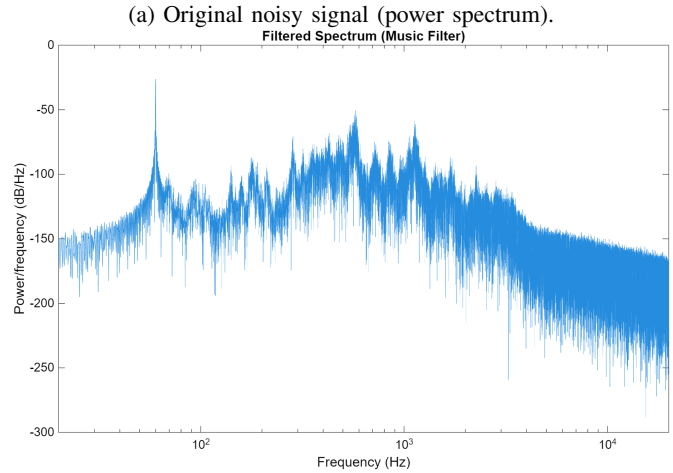
C. Task 4.3: Music Filter

Goal: Remove $60\,\mathrm{Hz}$ hum and high-frequency "hiss".

Design Justification: The analysis of the noisy (Fig. 7(a)) and clean audio showed that music exists between $\approx 200\,\mathrm{Hz}$ and $2000\,\mathrm{Hz}$, while noise exists at $60\,\mathrm{Hz}$ and $> 2000\,\mathrm{Hz}$. A bandpass filter with $f_0 \approx 632\,\mathrm{Hz}$ and $\Delta f = 1800\,\mathrm{Hz}$ was designed. We chose $C = 0.1\,\mu\mathrm{F}$, $L \approx 0.63\,\mathrm{H}$, and $R \approx 7.1\,\mathrm{k}\Omega$. **Result:** The filter performed exceptionally well. As shown in Fig. 7(b), the $60\,\mathrm{Hz}$ hum and high-frequency hiss are mostly eliminated. The Bode plot in Fig. 8 confirms the design, showing deep attenuation at $60\,\mathrm{Hz}$ and above $2000\,\mathrm{Hz}$.



(a) Original noisy signal (power spectrum).



(b) Filtered signal (power spectrum).

Fig. 7: Power spectra for Music Filter.

VII. Conclusion

This case study successfully demonstrated the derivation and application of a discrete-time state-space model for RLC circuits. We analyzed the model's behavior, paying close attention to the critical relationship between R and h in determining numerical stability, a key artifact of the Forward Euler method. The analysis confirmed that the V_R output of a series RLC circuit functions as a bandpass filter, which we then exploited to design three high-performance filters for the competition.

A. Limitations and Future Work

The primary limitation is the Forward Euler method, chosen for its simplicity. Its conditional stability ($R > h/C$) is not physically accurate. Future work should implement a more robust, unconditionally stable technique, such as the Bilinear Transform (Tustin's Method), which correctly maps the stable s-plane to the inside of the z-plane unit circle. Additionally, the RLC circuit is a simple filter; more advanced digital IIR filters

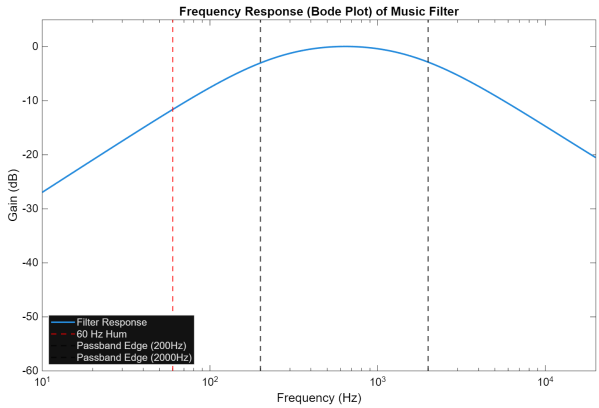


Fig. 8: Analytic Bode (magnitude) plot of the final music filter. The plot confirms the design, showing deep attenuation at 60 Hz (red line) and above 2000 Hz (right black line), with a clear passband in between.

(e.g., Butterworth, Chebyshev) could be explored for superior control over passband flatness and transition bandwidth [3].

Member Contributions

Derek Nester:

- Latex (Abstract, RC Circuit Analysis, RL Circuit Analysis, RLC Circuit Analysis)
- Matlab (Case Study Part 1, 2, and 3 and Competition Filter Design).

Riley Phil Panaligan:

- Latex (Introduction, Competition Filter Design, and Conclusion)

References

[1] ESE 105 Case Study 2 Instructions, "Circuits as Resonators, Sensors, and Filters," Washington University in St. Louis, Fall 2025.
[2] J. W. Nilsson and S. A. Riedel, *Electric Circuits*, 11th ed. Pearson, 2018.
[3] A. V. Oppenheim and A. S. Willsky, *Signals and Systems*, 2nd ed. Prentice-Hall, 1997.